Explainable Hybrid Detection of AI-Generated and Human-Written Content Using DistilBERT and Stylometric Machine Signatures

Your Name
Department of Artificial Intelligence
Your University
your.email@university.edu

Abstract—**The rapid growth of generative artificial intelligence models such as GPT, Gemini, and Claude has increased the need for reliable methods to distinguish between human-written and AI-generated content. This project addresses the binary classification problem of detecting AI-generated versus human-written text and code. A comparative analysis was conducted between traditional machine learning, deep learning, and a proposed hybrid model. The proposed model combines DistilBERT semantic embeddings with stylometric machine-signature features, including punctuation ratio, lexical diversity, repeated-word ratio, sentence length, and code-symbol ratio. To ensure a realistic evaluation, explicit leakage markers such as "AI-generated" were removed, and metadata fields such as source, prompt, and AI model name were excluded. The proposed hybrid model achieved an accuracy of 75.43%, balanced accuracy of 75.42%, macro-F1 score of 0.754, and ROC-AUC of 0.879. The results show that combining contextual transformer representations with stylometric features provides a balanced and explainable approach to AI-authorship detection.**

Index Terms—**AI-generated content, human writing detection, DistilBERT, stylometry, machine signatures, information integrity, AI safety**

I. Introduction

Generative artificial intelligence has become widely used for producing essays, reports, articles, and programming code. Modern large language models can generate fluent and coherent content that is increasingly difficult to distinguish from human-authored work. This creates challenges in academic integrity, misinformation detection, digital forensics, and information trust.

The objective of this project is to build a binary classifier that distinguishes between human-written and AI-generated content. Unlike simple detector models that rely only on word frequency or black-box neural predictions, this work focuses on explainable machine-signature analysis. The proposed approach combines semantic information learned by DistilBERT with stylometric features that describe measurable writing and code characteristics.

The main contribution of this project is a hybrid detection framework that integrates deep contextual embeddings with interpretable stylometric signals. This design allows the model not only to classify content, but also to provide insight into which writing patterns differ between AI-generated and human-written samples.

II. Related Work

Previous AI-generated text detection methods can be grouped into classical machine learning methods, transformer-based methods, and stylometric approaches. Classical models such as Logistic Regression and Support Vector Machines often use TF-IDF features to classify text based on lexical patterns. Although these approaches are fast and interpretable, they may rely heavily on surface-level cues.

Transformer-based models such as BERT, RoBERTa, and DistilBERT have shown strong performance in text classification tasks because they capture contextual and semantic relationships. A recent study on AI-generated scientific abstract detection showed that DistilBERT can achieve strong results while remaining computationally efficient. The study also emphasized the importance of cross-domain testing, leakage removal, and robustness evaluation.

Stylometric approaches focus on measurable writing patterns such as sentence length, vocabulary richness, punctuation usage, syntactic structure, and repetition. Recent literature suggests that AI-human distinction should not depend on one universal marker, but rather on multiple cue families, including surface cues, discourse cues, probabilistic cues, and content-level cues. This supports the motivation for using a hybrid model that combines semantic and stylometric features.

III. Methodology

A. Dataset

The dataset used in this project contains 20,000 samples of human-written and AI-generated content. The class distribution is approximately balanced, with 10,004 AI-generated samples and 9,996 human-written samples. The dataset includes both text and code samples, making the task more realistic and challenging.

B. Data Cleaning and Leakage Removal

During the dataset audit, an important leakage issue was identified: all AI-generated samples contained the explicit phrase "AI-generated". If left unchanged, the model could learn this phrase directly instead of learning real authorship patterns. Therefore, this marker was removed from all samples before training.

Other preprocessing steps included removing URLs, email-like patterns, excessive whitespace, and very short samples. Metadata columns such as source, prompt, AI model name, and content type were excluded from the model input to prevent shortcut learning.

C. Stylometric Feature Extraction

Stylometric features were extracted from each content sample to represent interpretable machine-signature characteristics. The extracted features included:

- Character count
- Word count
- Sentence count
- Average word length
- Average sentence length
- Type-token ratio
- Repeated-word ratio
- Punctuation ratio
- Digit ratio
- Uppercase ratio
- Code-symbol ratio
- Quote count
- Bracket count
- Semicolon count

These features were selected because they capture differences in writing regularity, lexical diversity, punctuation behavior, and code structure.

D. Proposed Hybrid Model

The proposed model combines DistilBERT embeddings with stylometric features. First, each input sample is tokenized using the DistilBERT tokenizer. DistilBERT then generates contextual token embeddings. Instead of relying only on the first token representation, mean pooling is applied across token embeddings using the attention mask.

The stylometric feature vector is normalized and passed through a small projection layer. The DistilBERT representation and stylometric representation are then concatenated and passed into a neural classification head.

The architecture can be summarized as follows:

$$\text{Input Content} \rightarrow \text{DistilBERT Embedding}$$

$$\text{Input Content} \rightarrow \text{Stylometric Features}$$

$$\text{Combined Representation} \rightarrow \text{Classifier} \rightarrow \text{Human or AI}$$

E. Training Setup

The dataset was divided into training, validation, and testing sets. The split sizes were:

- Training set: 12,800 samples
- Validation set: 3,200 samples
- Test set: 4,000 samples

The model was trained using the AdamW optimizer with a learning rate of 2×10^{-5}. Binary cross-entropy with logits loss was used as the loss function. The model was trained for three epochs, and the best model was selected based on validation macro-F1 score. A validation-based threshold was also selected to avoid over-predicting one class.

IV. Results and Discussion

A. Validation Results

The best model was selected from epoch 2 based on validation macro-F1 score. The validation results were:

- Validation accuracy: 74.56%
- Validation balanced accuracy: 74.56%
- Validation macro-F1: 0.7449
- Validation ROC-AUC: 0.8711

Although epoch 3 achieved slightly higher validation accuracy, it had a lower macro-F1 score. Therefore, epoch 2 was selected because macro-F1 better reflects balanced performance across both human and AI classes.

B. Final Test Results

The final test results are shown in Table I.

TABLE I
FINAL TEST PERFORMANCE OF THE PROPOSED HYBRID MODEL

Metric	Value
Accuracy	75.43%
Balanced Accuracy	75.42%
AI F1-score	0.764
Macro-F1	0.754
ROC-AUC	0.879

The confusion matrix is shown in Table II.

TABLE II
CONFUSION MATRIX

	Predicted Human	Predicted AI
Actual Human	1427	572
Actual AI	411	1590

The classification report showed that the model achieved 0.78 precision and 0.71 recall for the human class, and 0.74 precision and 0.79 recall for the AI class. This indicates that the improved model produced a more balanced classification behavior compared with the earlier version, which over-predicted AI-generated content.

C. Comparative Analysis

The comparative analysis considered three model categories: classical machine learning, deep learning, and the proposed hybrid model. Classical TF-IDF-based models provide interpretability but are limited in semantic understanding. DistilBERT-only models capture deeper contextual patterns but provide limited explicit explanation of machine signatures. The proposed hybrid model combines both strengths by using DistilBERT for semantic representation and stylometric features for explainability.

The proposed model was selected as the final model because it provides the best balance between predictive performance and interpretability. Unlike a black-box transformer-only approach, the hybrid model supports analysis of measurable writing patterns.

D. Stylometric Machine-Signature Analysis

The stylometric analysis showed that AI-generated samples had slightly higher average character count, word count, punctuation ratio, repeated-word ratio, and code-symbol ratio. Human-written samples showed slightly higher type-token ratio, suggesting marginally greater lexical diversity.

These differences were small but meaningful when combined with DistilBERT embeddings. This supports the idea that AI-generated content does not always contain a single obvious signature. Instead, detection depends on combinations of subtle semantic and stylistic cues.

The higher repeated-word ratio in AI-generated content may indicate more regular or formulaic phrasing. The slightly lower type-token ratio suggests that AI-generated samples may use a narrower vocabulary distribution compared with human-written samples. These observations support the project focus on machine-signature analysis.

V. Conclusion

This project presented an explainable hybrid model for detecting AI-generated versus human-written content. The proposed system combined DistilBERT contextual embeddings with stylometric features representing writing and code-level machine signatures. The dataset was carefully cleaned to remove explicit leakage markers and metadata shortcuts, making the evaluation more realistic.

The final model achieved 75.43% accuracy, 0.754 macro-F1, and 0.879 ROC-AUC on the test set. More importantly, the model achieved balanced performance across both human and AI classes, avoiding the earlier issue of over-predicting AI-generated content.

The results show that hybrid semantic-stylometric modeling is a promising approach for AI-authorship detection. Future work can improve the system by testing generalization across unseen AI models, evaluating robustness against paraphrasing attacks, and applying explainability methods such as SHAP or attention visualization.

References

[1] C. Zhang and W. Zhou, “Efficient detection of AI-generated scientific abstracts with a lightweight transformer,” *Scientific Reports*, 2026.

[2] G. P. Georgiou, “What distinguishes AI-generated from human writing? A rapid review of the literature,” *Big Data and Cognitive Computing*, vol. 10, no. 55, 2026.

[3] J. Devlin, M. Chang, K. Lee, and K. Toutanova, “BERT: Pre-training of deep bidirectional transformers for language understanding,” in *Proc. NAACL-HLT*, 2019.

[4] V. Sanh, L. Debut, J. Chaumond, and T. Wolf, “DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter,” 2019.